Calcurion Maths Worksheets

A-Level Mathematics

Topic: Statistics – Hypothesis Testing

Medium & Hard • 25 Questions

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Section A – Medium (Q1–15)

1. State the null and alternative hypotheses for testing whether a coin is biased towards heads. **[2]**

2. Define a significance level. **[2]**

3. What is meant by a Type I error? **[2]**

4. What is meant by a Type II error? **[2]**

5. $X \sim B(10, 0.5)$. For $H_0 : p = 0.5$ vs $H_1 : p > 0.5$ at 5% level, find the critical region. **[4]**

6. A test is conducted at the 5% significance level. What does this mean? **[2]**

7. $X \sim B(20, p)$. $H_0 : p = 0.4$, $H_1 : p \neq 0.4$. Describe a two-tailed test. **[3]**

8. Observed value $x = 8$ from $B(12, 0.5)$. Test $H_0 : p = 0.5$ vs $H_1 : p > 0.5$ at 5%. **[5]**

9. State the conditions under which a normal approximation to the binomial may be used for a hypothesis test. **[3]**

10. Explain the difference between a one-tailed and a two-tailed test. **[3]**

11. $H_0 : \mu = 50$, $H_1 : \mu < 50$. A sample gives a test statistic leading to p-value $= 0.03$. Interpret at 5% level. **[3]**

12. Write the hypotheses for testing whether the mean of a normal population is greater than 100. **[2]**

13. A binomial test has critical region $X \leq 2$ when $n = 15$, $p = 0.3$. Find the actual significance level. **[4]**

14. Why do we never "accept H_0" but rather "fail to reject H_0"? **[2]**

15. A sample of size 20 from $N(\mu, 16)$ has mean 48. Test $H_0 : \mu = 50$ vs $H_1 : \mu < 50$ at 5%. **[5]**

Section B – Hard (Q16–25)

16. $X \sim B(25, p)$. Test $H_0 : p = 0.6$ vs $H_1 : p < 0.6$ at 5%. Observed $x = 10$. Carry out the test. **[6]**

17. Find the critical region for $X \sim B(12, 0.4)$ testing $H_0 : p = 0.4$ vs $H_1 : p > 0.4$ at the 5% level. **[5]**

18. A machine fills bags with mean mass 500 g and SD 5 g (normal). A sample of 25 bags has mean 498 g. Test at 5% whether the mean has decreased. **[6]**

19. Explain what a p-value is and how it is used in hypothesis testing. **[3]**

20. $X \sim B(40, p)$. Using a normal approximation, test $H_0 : p = 0.5$ vs $H_1 : p \neq 0.5$ given $\bar{x}$ such that 28 successes observed, at 5%. **[6]**

21. The critical region for a test is $X \leq 3$ or $X \geq 12$ when $n = 15$. State the hypotheses and find the significance level if $p = 0.5$ under H_0. **[5]**

22. A company claims that 90% of its products meet a standard. In a sample of 50, 40 meet the standard. Test the claim at 5% (one-tailed). **[6]**

23. Distinguish between the critical region approach and the p-value approach. **[3]**

24. Sample mean from $N(\mu, 9)$ with $n = 36$ is 22. Test $H_0 : \mu = 20$ vs $H_1 : \mu > 20$ at 1% level. **[5]**

25. Why is the significance level chosen before collecting data? **[2]**

End of Worksheet • *Show all working clearly.*